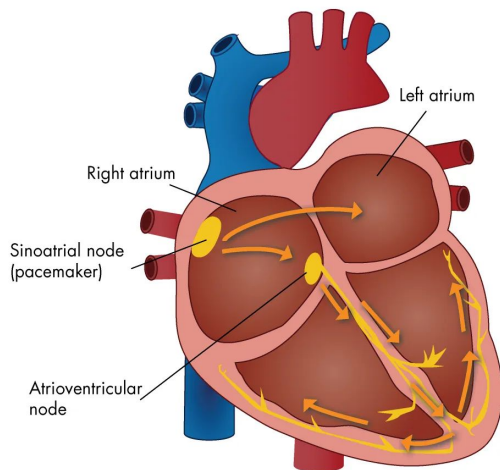


Carbon Nanotube (CNT)-Doped Elastomeric Round Tip Catheter with Force Feedback For Cardiac Ablation Procedures

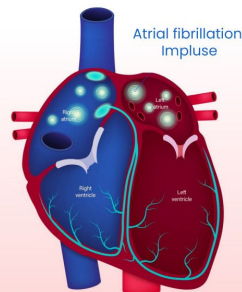
Abhishek Lele

Content Overview

- 1 Introduction/Background
- 2 Limitations/Problem Statement
- 3 Proposed Solution
- 4 Innovation of Solution
- 5 Impact Discussion

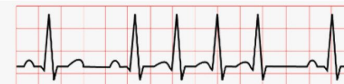


Arrhythmias

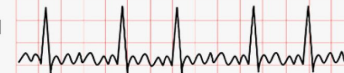


Arrhythmia classification

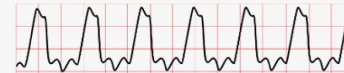
Atrial fibrillation



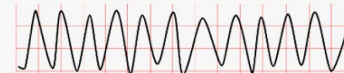
Paroxysmal atrial tachycardia



Ventricular tachycardia



Ventricular fibrillation

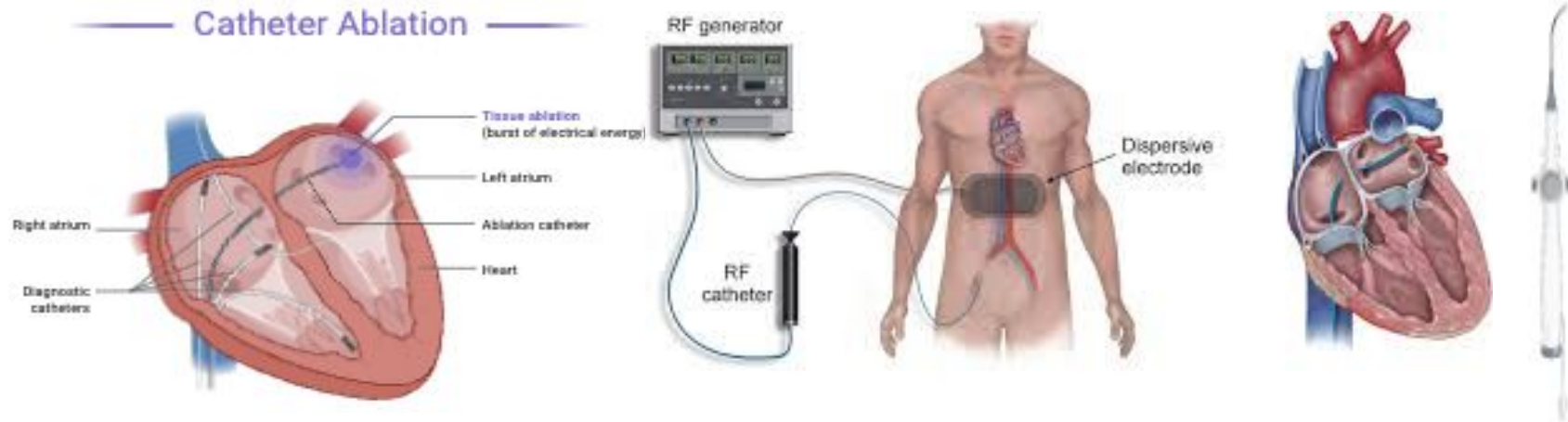


If Parts of the heart are unable to conduct properly, the signal becomes inconsistent

What is Cardiac Ablation?

Cardiac ablation is a minimally invasive procedure that treats arrhythmias by creating small, targeted scars in the heart to block faulty electrical signals.

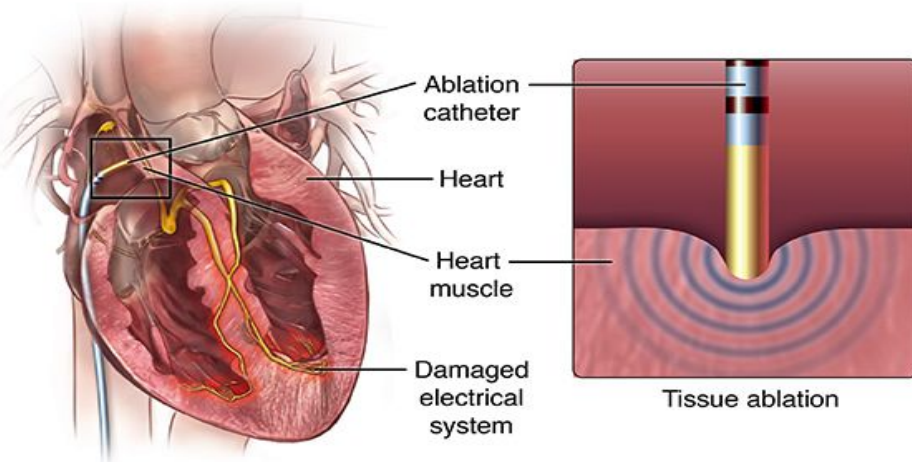
Thin catheter through groin or neck → Guided to heart through mapping + X-ray imaging →
Positioned at abnormal electrical site → Apply cryo energy to targeted tissue → rhythm restored



Why do Contact Forces Matter?

A significant number of complications involve unintended damage caused by the catheter

Catheter ablation



RISKS

- Vascular Tissue damage
- Bleeding at catheter insertion site
- Nerve Damage
- Puncture to the heart wall
- Inconsistent Catheter Tissue contact force

300,000+
surgeries every
year in US



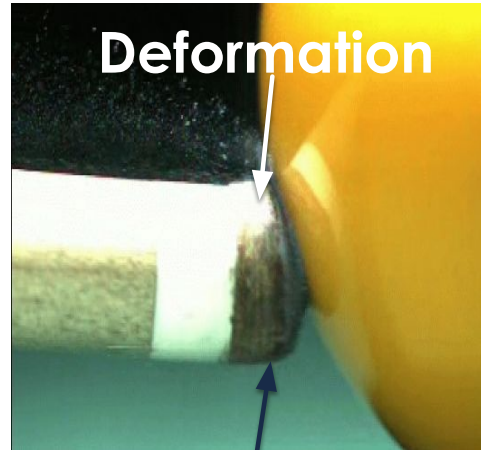
~ 15,000
involve
complications

Force mapping can solve our issues!

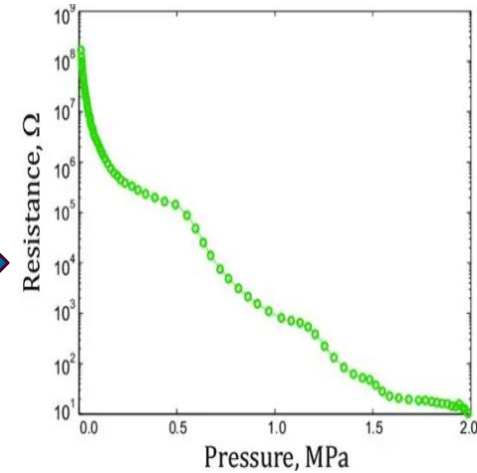
Piezoresistive Material

- Electrical resistance of a material (typically a semiconductor) changes when mechanical strain is applied.
Used in sensors for Measuring
- Pressure, force, and acceleration

We plan to implement these properties in a biocompatible soft resin tip with a biocompatible CNT doping procedure.



Piezoresistive tip



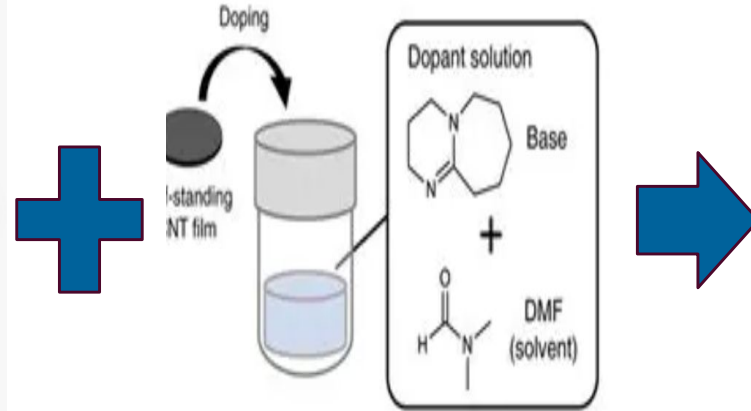
Results in change in resistance of material

Tip Mechanism (Project Plan)



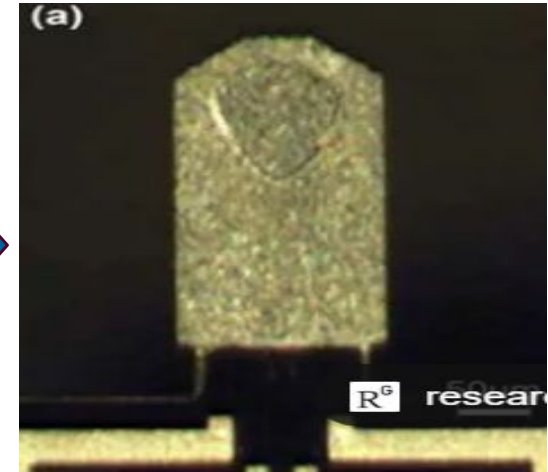
Elastomeric tip

soft, compliant distal catheter component



Biocompatible CNT doping

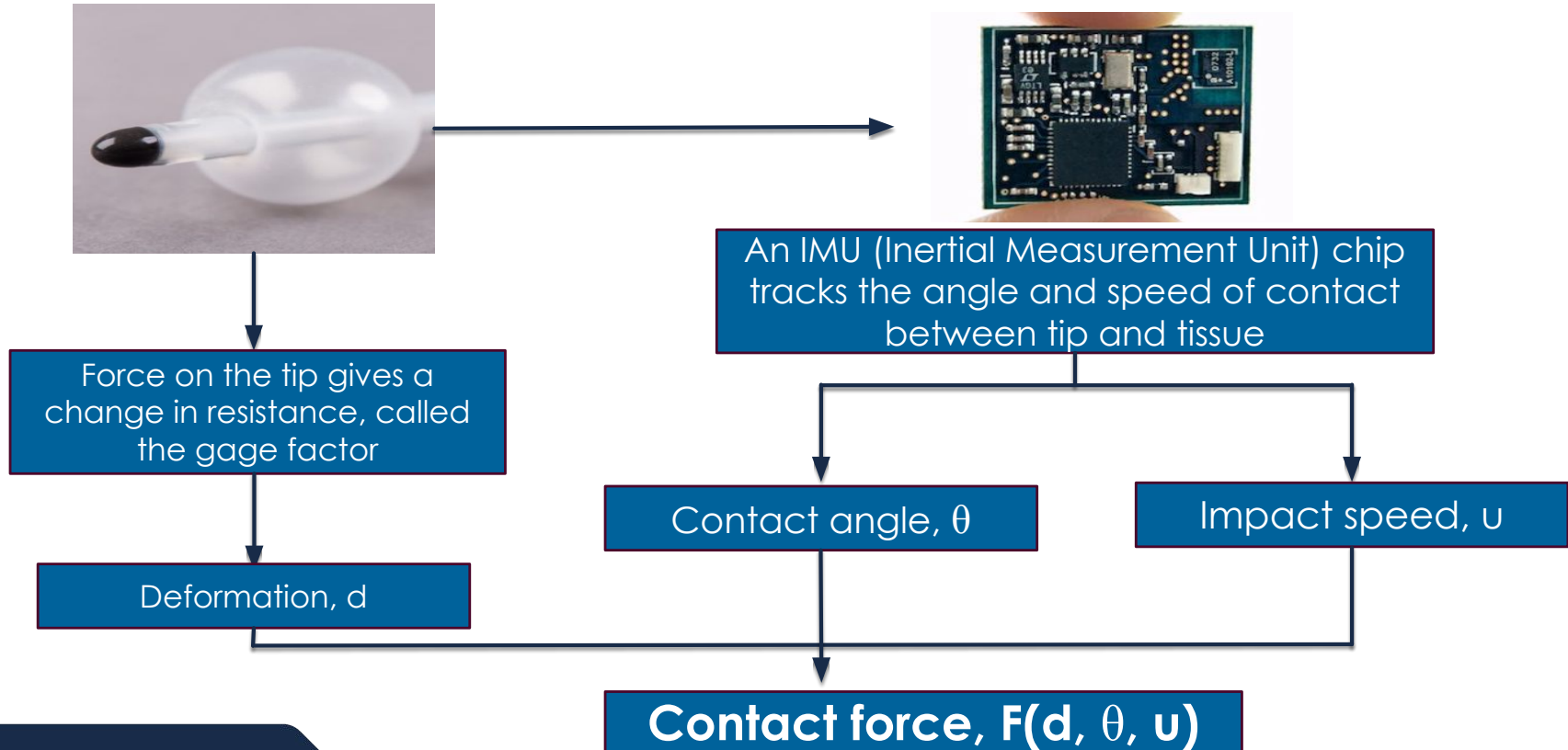
embedding Carbon Nanotubes to impart electrical conductivity and piezoresistive properties



Piezoresistive CNT-doped tip

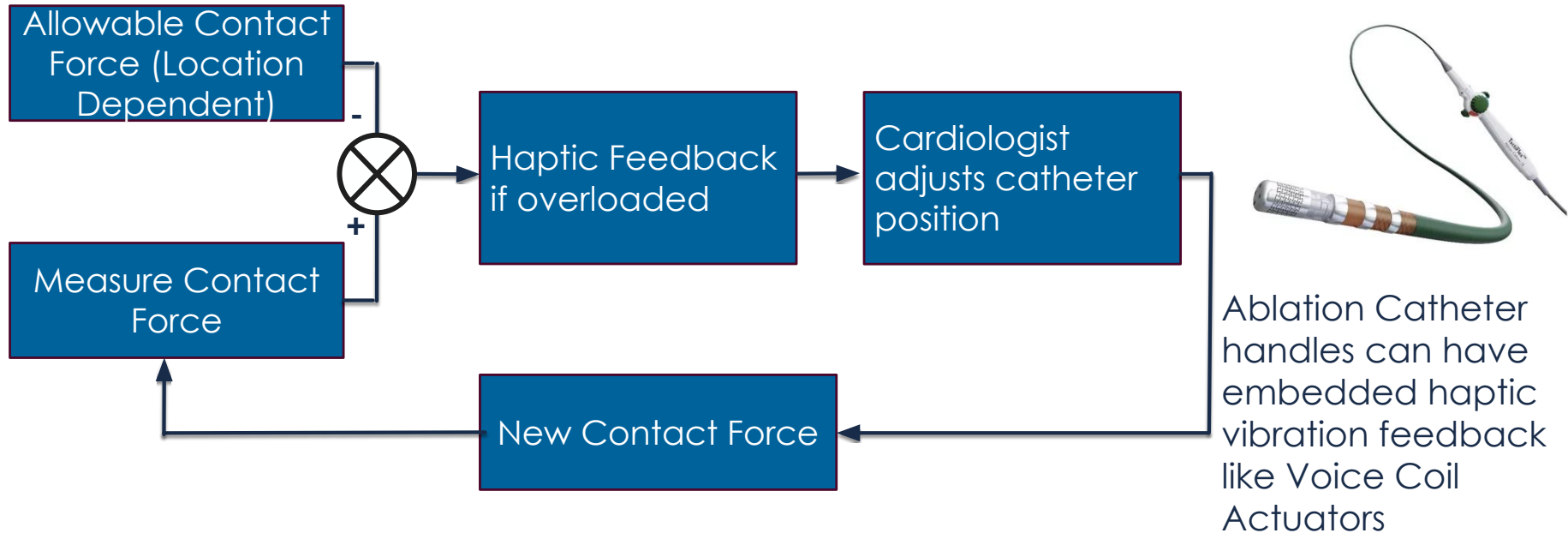
converts mechanical deformation into measurable resistance changes, for real-time contact force sensing

Force Measurement Design



Full Workflow

Human Controlled Closed-Loop System:



Aims

Aim #1: Material Evaluation

Evaluate the biocompatibility and material safety of the CNT-doped elastomeric tip based on ISO standards and previous literature.

Considerations:

- cytotoxicity
- surface interactions
- compliance with relevant standards for safe clinical use

Aim #2: Tip Design

Test three different biocompatible resins using Finite Element Analysis (FEA) to evaluate stress reactions with different parts of the heart and find the best structural and strength fit.

<u>Material</u>	<u>Mechanical Properties</u>
Medical-grade silicone	Low Young's modulus, great flexibility under cyclic loading
TPU (Thermoplastic Polyurethane)	Moderate modulus, elastic recovery after deformation
Pebax-Based Elastomer	Tuneable stiffness good flexibility, high fatigue resistance

Table: Materials Considered for Testing and Reasoning

Aim #3: Force Model

Design a real time predictive model to estimate contact force based on inputs such as

RESISTANCE

CONTACT VELOCITY

ANGLE

This model will enable accurate, closed-loop force sensing for improved procedural control.

Aim #3: Force Model

A 2D finite element model of the CNT-doped elastomeric catheter tip will be developed to calculate **Contact force, $F(d, \theta, u)$** .

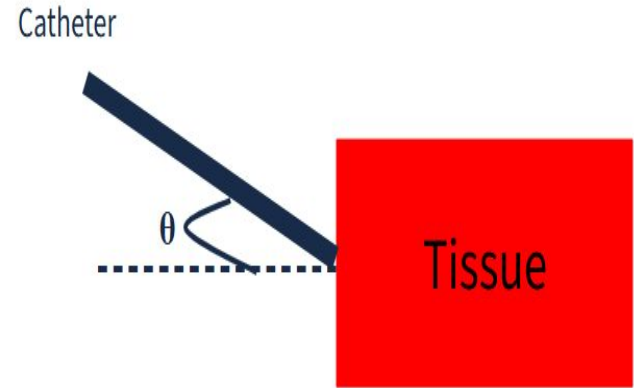
Variables include:

1. Material comparison
2. Different contact speeds and angles

Further, study different doping methods focused on optimizing CNT concentrations to utilize piezoresistive properties.

Further animal (mouse) and human testing will be conducted to ¹test the system with the body's electrophysiological response.

Safe Force:
~0.1-0.4N



Simulation of Proposed FEA Modelling

CNT-doped force sensing enables a biocompatible and safer, data-driven, and potentially autonomous catheter navigation with reduced reliance on imaging

Advantages Over Current Systems

Advantage	Impact Compared to Conventional Tips
Real-time Force Feedback	Quantitative contact force at the tip instead of relying on physician intuition
Improved Patient Safety	Reduces risk of tissue perforation, excessive strain, and unintended damage
Reduced Imaging Dependency	Minimizes reliance on fluoroscopy/X-ray, lowering radiation exposure
Enhanced Durability (CNT Doping)	Embedded CNT network withstands large deformation better than fragile surface layer approaches

References

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- [2. An advanced robotic system incorporating haptic feedback for precision cardiac ablation procedures – PMC](#)
- [3. Emerging Tools and Techniques for Catheter Ablation of Cardiac Arrhythmias: A 2024 Update – PMC](#)
- [4. A Novel Catheter Distal Contact Force Sensing for Cardiac Ablation Based on Fiber Bragg Grating with Temperature Compensation – PMC](#)
- [5. Contact force mapping during catheter ablation for atrial fibrillation: procedural data and one-year follow-up – PMC](#)
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Thank you!

